Report Of Flying Whale Airline

Business intelligence case study

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Introduction

FlyingWhale Airline, a renowned (fictional) international airline, has embarked on an initiative to enhance its business intelligence capabilities. This report provides insights derived from the analysis of Customer Flight Activity and Customer Loyalty History, with the aim of optimizing customer experience, understanding travel patterns, and maximizing loyalty program effectiveness.

Background

FlyingWhale Airline, a prominent (fictional) international airline, is seeking to enhance its business intelligence capabilities by analyzing Customer Flight Activity and Customer Loyalty History. The airline is committed to optimizing customer experience, understanding travel patterns, and maximizing the effectiveness of its loyalty programs.

Data Cleaning

- Verified completeness and consistency of initial datasets.
- Cleansed data to correct inconsistencies and remove duplicates.
- Standardized data formats for integration and comparison.
- Enriched data with additional information for a comprehensive view.
- Detected and addressed anomalies or outliers.
- Normalized data to a common scale for comparative analysis.

Key Performance Indicators

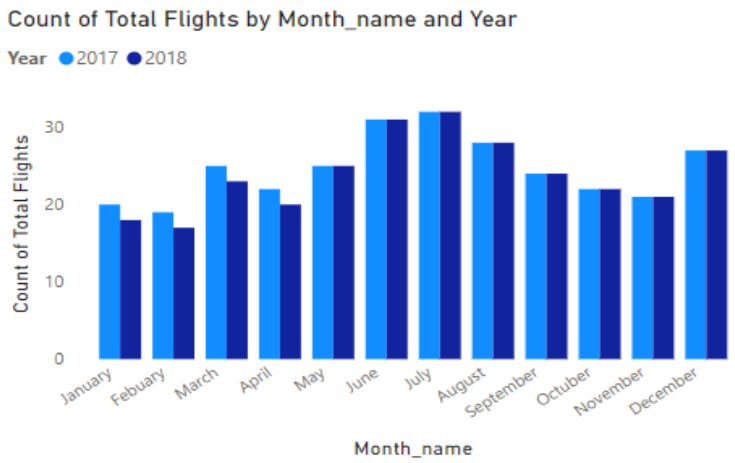
- **Total Flights Booked**: 2 million
- **Total Members**: 16.74 thousand
- **Total Distance Covered**: 490 million Miles
- **Total Points Accumulated**: 50.17 million
- **Total Points Redeemed**: 12 million

Flight Activity Analysis

This analysis of the flight activity for Flying Whale Airline, focusing on the years 2017 and 2018. The data has been meticulously gathered and visualized to provide insights into monthly flight counts, the impact of flights with companions on loyalty points redemption, and the correlation between flights booked and distance covered.

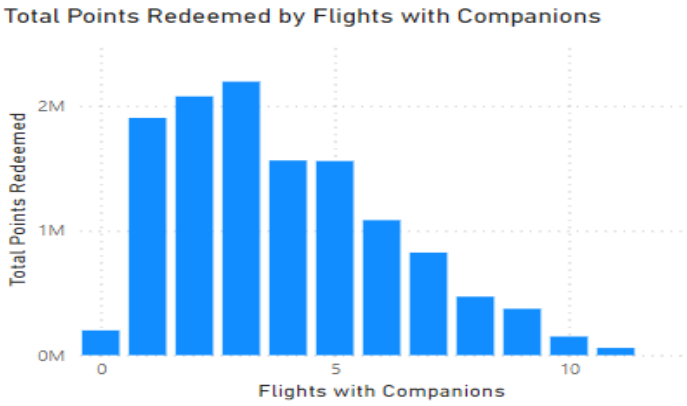
Monthly Flight Counts:

- The bar graph comparison of total flights by month for 2017 and 2018 shows fluctuating patterns with peaks in March, May, July, and October, indicating increased flight activities during these periods.



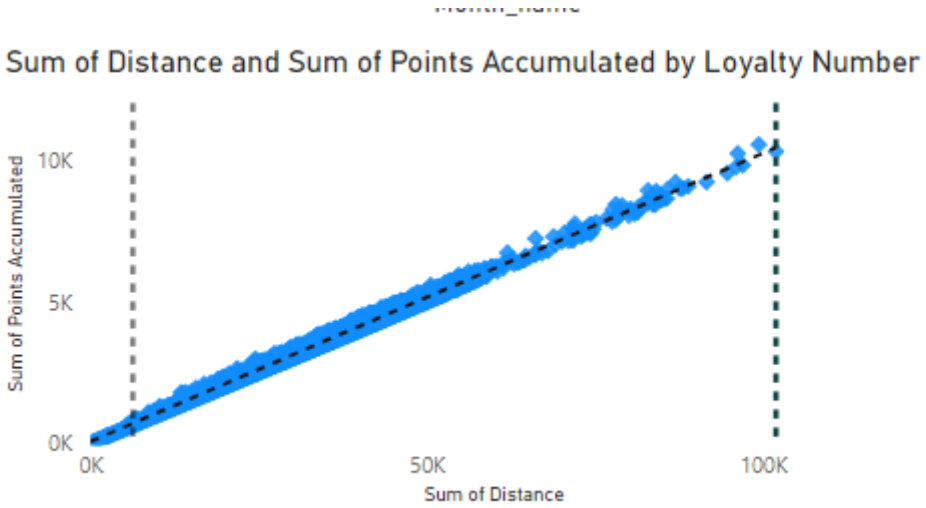
Points Redeemed by Flights with Companions:

- The analysis reveals a decline in total points redeemed as the number of flights with companions increases, with maximum redemptions observed for solo flights or those with fewer companions.



Correlation between Flights Booked and Distance Covered:

- A scatter plot analysis shows a strong positive correlation of 0.95, indicating that as more flights are booked, the cumulative distance covered also increases proportionately.



Findings of Flight Activity Analysis

The flight activity analysis has highlighted several key findings:

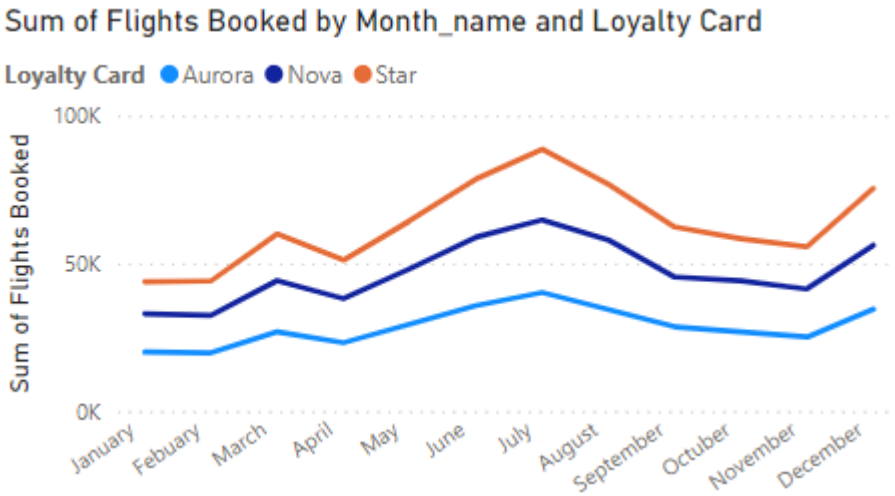
- There is a seasonal pattern in flight bookings, with specific months experiencing higher activity.
- Customers tend to redeem fewer points when traveling with companions.
- A strong positive correlation exists between the number of flights booked and the distance covered, suggesting that loyalty programs are effectively encouraging frequent flying.

Loyalty Segmentation

The Loyalty Segmentation Analysis for Flying Whale Airline has provided valuable insights into the behaviors and preferences of loyalty cardholders. These findings will inform targeted marketing strategies and loyalty program enhancements to improve customer retention and increase CLV.

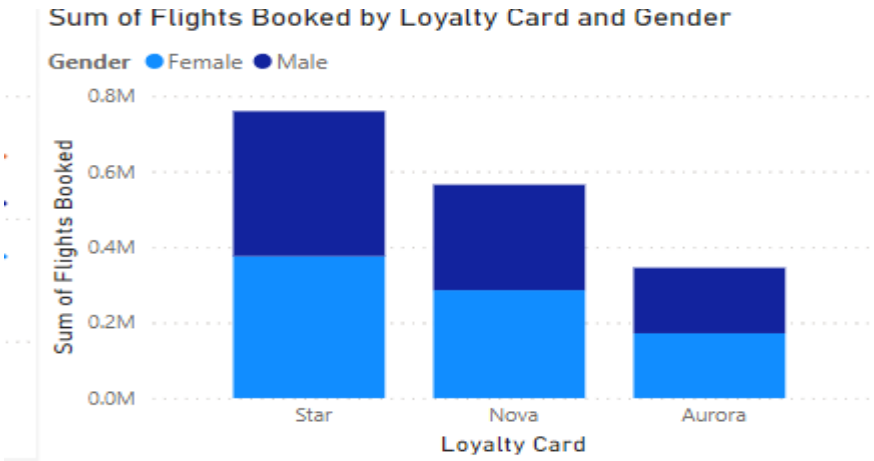
Monthly Bookings Analysis:

- The line graph illustrates the sum of flights booked per month, segmented by loyalty card types. A consistent booking pattern is observed across all categories, with fluctuations indicative of seasonal trends.



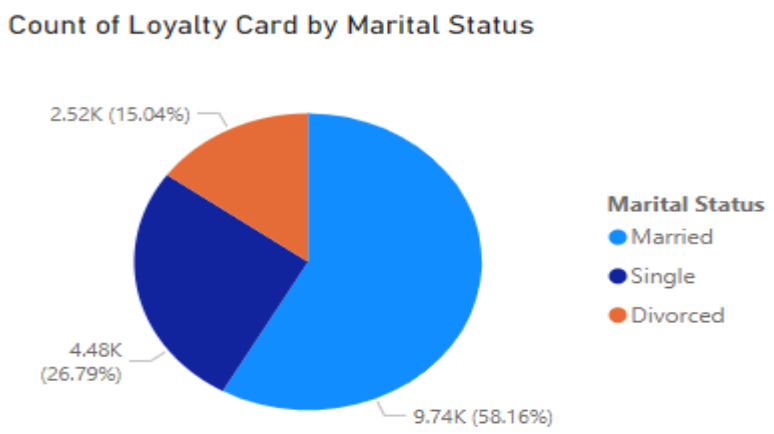
Gender-based Bookings:

- The stacked bar chart reveals booking data for male and female passengers holding different types of loyalty cards. Males predominantly hold Star cards, while females are more inclined towards Nova cards.



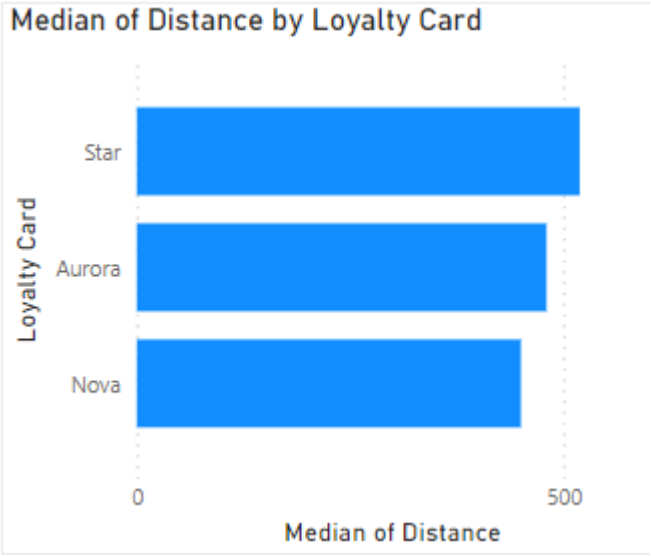
Marital Status Distribution:

- The pie chart displays the distribution of loyalty cards based on marital status, showing a significant proportion of cardholders are married.



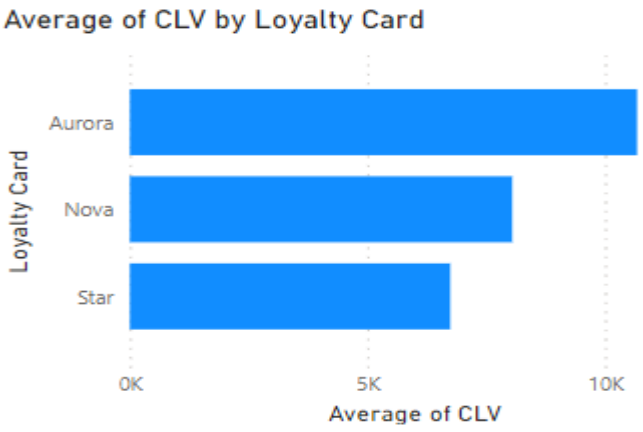
Distance Traveled Analysis:

- The horizontal bar chart showcases the median distance traveled by holders of different loyalty cards, with Star cardholders traveling shorter distances on average.



Customer Lifetime Value (CLV):

- The horizontal bar chart represents the average CLV for each type of loyalty card, with Aurora cardholders exhibiting a higher CLV compared to Nova and Star members.



Findings Of Loyalty Segmentation

The loyalty segmentation analysis has highlighted several key findings:

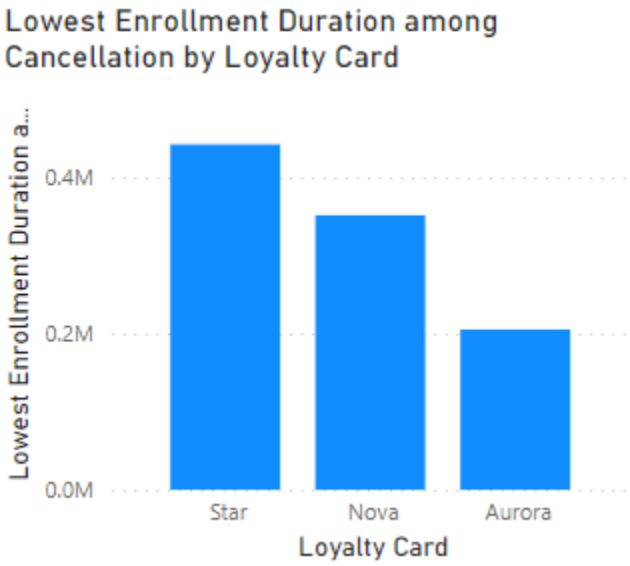
- Seasonal trends affect the booking patterns across all loyalty card types.
- Gender distribution varies significantly among different loyalty cardholders.
- Married customers form a substantial portion of the loyalty program.
- There is a variation in travel behavior based on the type of loyalty card held.
- Aurora cardholders have the highest average CLV, indicating higher engagement and value.

Enrollment and Cancellation Trends

It examines the enrollment and cancellation trends within the loyalty program of Flying Whale Airline. The analysis is segmented by loyalty card type, month, education, marital status, and province. The findings are visualized through bar graphs to provide clear insights into the factors influencing loyalty program dynamics.

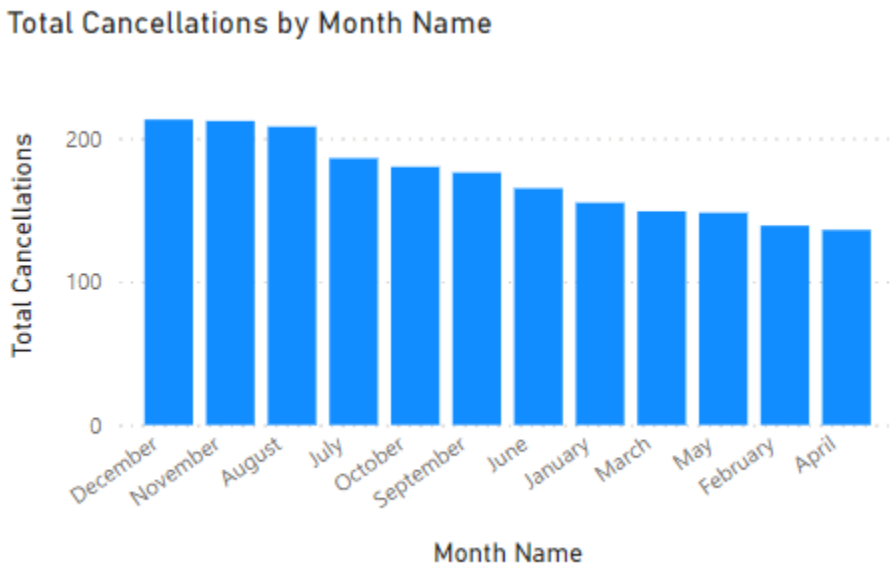
Loyalty Card Duration:

- Members with the Aurora loyalty card have the highest average duration of days enrolled, indicating strong loyalty and satisfaction with the program.



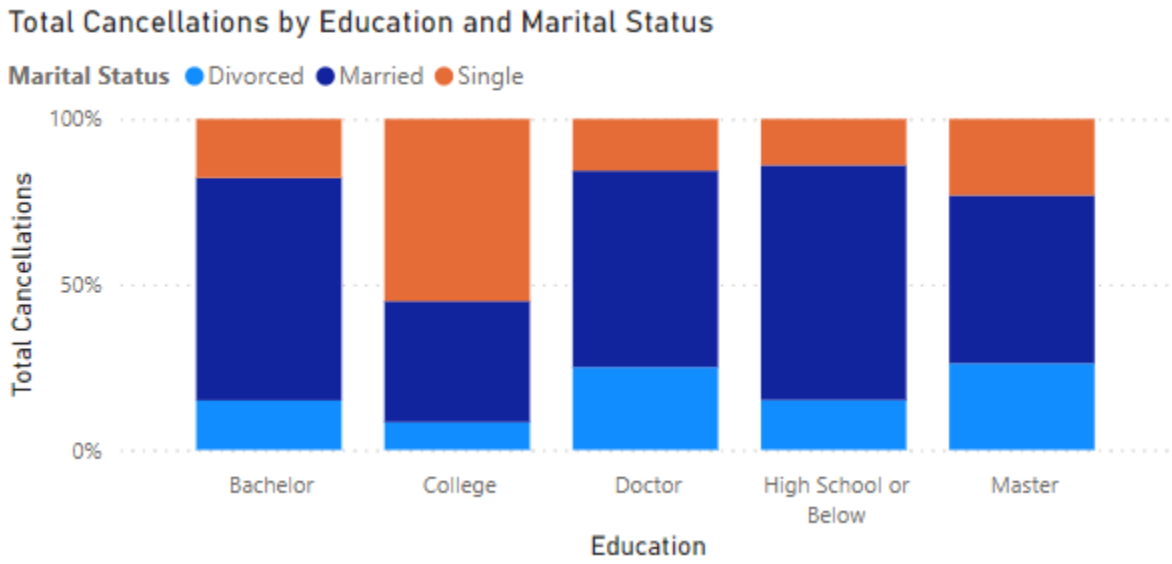
Monthly Cancellation Trends:

- Cancellations peak in December, suggesting a possible seasonal influence on members' decisions to cancel their loyalty memberships



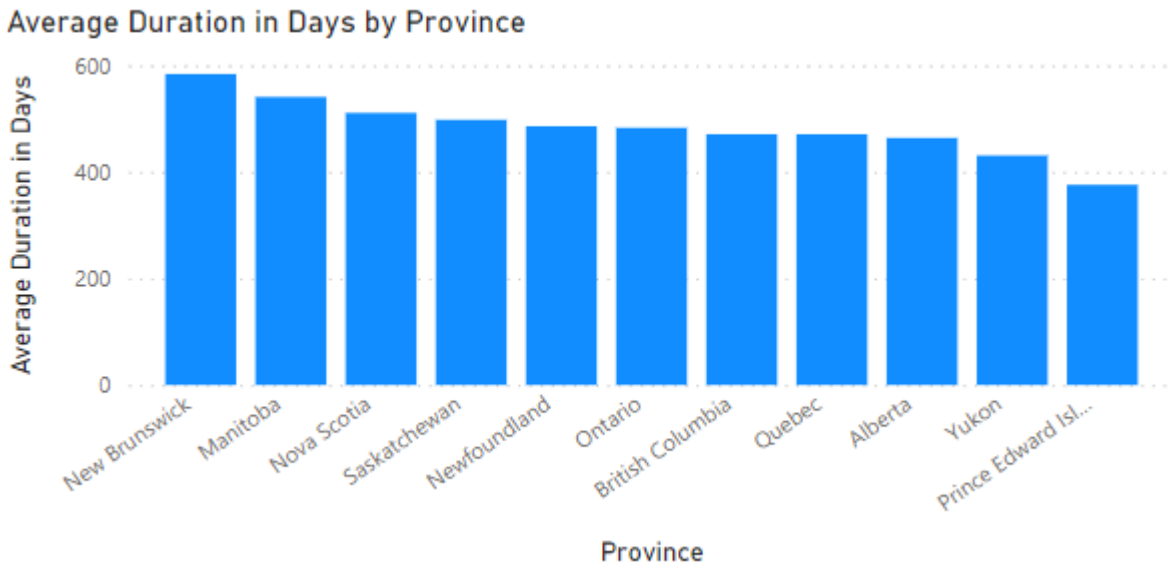
Education & Marital Status Impact on Cancellations:

- Individuals with a bachelor's degree represent the highest number of cancellations, with married individuals forming a significant proportion of this group.



Provincial Enrollment Duration:

- New Brunswick leads in average enrollment duration, suggesting higher satisfaction or better engagement strategies in this region.



Findings

- The analysis of enrollment and cancellation trends has revealed several key insights:
- Loyalty card type is a significant factor in the duration of membership.
- There is a clear seasonal pattern in cancellation rates, with a spike in December.
- Educational background and marital status are correlated with cancellation rates.
- Provincial differences in enrollment duration may indicate varying levels of engagement or program effectiveness.

Recommendation

Following are the strategies for improving enrollment and retention for Flying Whale Airline,

1. **Launch Seasonal Loyalty Programs**: Introduce special benefits during the months with high cancellation rates to retain members.
2. **Customize Communication**: Tailor messages for demographics with high cancellation rates, such as bachelor's degree holders and married individuals.
3. **Reward Longevity**: Provide additional perks or status upgrades to long-term members to recognize and encourage continued loyalty.
4. **Regional Program Adaptation**: Adjust loyalty benefits to cater to the preferences of members in provinces with varying enrollment durations.
5. **Offer Educational Incentives**: Create incentives for college-educated and single individuals to increase their engagement with the loyalty program.
6. **Conduct Awareness Campaigns**: Promote the loyalty program's benefits to boost enrollment and educate members about its value.
7. **Implement a Feedback System**: Collect and act on feedback from members, especially those who cancel, to improve the loyalty program.
8. **Use Predictive Analytics**: Employ data analytics to identify members at risk of cancellation and proactively address their concerns.

By focusing on these strategies, Flying Whale Airline can enhance its loyalty program's effectiveness and improve both enrollment and retention rates.

Conclusion

In conclusion, the analysis of Flying Whale Airline's enrollment and cancellation trends has provided valuable insights into the loyalty program's performance. The strategies outlined aim to address the identified patterns and demographic factors influencing member retention. By implementing targeted engagement programs, personalized communication, and regional customization, along with leveraging feedback and predictive analytics, the airline can enhance the effectiveness of its loyalty program.